

SKILLS

PROGRAMMING

C++ • Python • CUDA • CMake

FRAMEWORKS & TOOLS

ROS/ROS2 • PyTorch • OpenCV • Git • Docker

RESEARCH

SLAM • 3D LiDAR • Point Cloud Registration • Ground Segmentation • Semantic Segmentation • Sensor Fusion • Deep Learning • Multi-Robot Mapping

LANGUAGES

Korean (Native) • English (Fluent)

LINKS

Website:// shaffar.com

GitHub:// [awshaff](https://github.com/awshaff)

LinkedIn:// [awachid](https://www.linkedin.com/in/awachid)

Scholar:// [Abdurrahman Wachid Shaffar](https://scholar.google.com/citations?user=AbdurrahmanWachidShaffar)

KEY OPEN-SOURCE PROJECTS

Projects others rely on, reflecting both coding quality and real-world impact.

- **KISS-Matcher** [🔗 ★ 643]
Out-of-the-box fast and robust global point cloud registration for large-scale mapping. (ICRA'25)
- **VGGT-SLAM** [🔗 ★ 882]
Dense real-time RGB SLAM on SL(4) manifold; globally consistent 3D reconstruction. (NeurIPS'25)
- **Patchwork++** [🔗 ★ 842]
>100 Hz LiDAR ground segmentation; deployed in AV and mobile robot production. (IROS'22)
- **ERASOR** [🔗 ★ 494]
~94% dynamic-object artifact removal, enabling long-term autonomous navigation. (RA-L'21)

EDUCATION

KYUNG HEE UNIVERSITY (KHU)

PH.D. IN ELECTRONICS AND INFORMATION CONVERGENCE ENGINEERING PROGRAM

Mar. 2022 -- Feb. 2028 | Yongin, South Korea

Advisor: Prof. Hyundong Shin

Dissertation: Quantum Microwave Communication

UNIVERSITAS GADJAH MADA (UGM)

B.S. IN PHYSICS

Aug, 2015 -- Nov. 2019 | Yogyakarta, Indonesia

Advisor: Prof. Pekik Nurwantoro

Dissertation: Quantum Mechanics

Implementation in Quantum Cryptography

EXPERIENCE

SPARK LAB, MIT | POSTDOCTORAL ASSOCIATE

Apr. 2024 -- Present | Cambridge, MA

- Reduced multi-session map inconsistency in city-scale environments by building a large-scale multi-robot SLAM system enabling real-time 3D mapping across 10+ robots and sessions.
- Accelerated research output by mentoring 4+ graduate researchers, contributing to 4+ published papers on 3D perception and autonomous mapping.
- Grew combined GitHub adoption to 4K+ stars by maintaining TEASER++ [🔗 ★ 2.1K], Hydra [🔗 ★ 773], KISS-Matcher [🔗 ★ 643], VGGT-SLAM [🔗 ★ 882].

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST) | POSTDOCTORAL FELLOW

Mar. 2023 -- Mar. 2024 | Daejeon, South Korea

- Improved localization robustness in dynamic scenes by developing real-time LiDAR-inertial and visual SLAM pipelines, deployed on 3+ mobile robot platforms.
- Reduced dynamic-object-induced map corruption by developing deep learning perception modules for moving object segmentation in crowded and unstructured environments.

STACHNISSLAB, UNIVERSITY OF BONN | VISITING SCHOLAR

Nov. 2022 -- Feb. 2023 | Bonn, Germany

- Cut dynamic-object artifacts in LiDAR maps by ~90% by building ERASOR2, an instance-aware removal system for static map building in urban scenes (RSS 2023).

NAVER LABS | RESEARCH INTERN

Apr. 2022 -- Sep. 2022 | Gyeonggi-Do, Korea

- Achieved robust global LiDAR registration tolerating up to 99% outlier correspondences by developing Quatro, exploiting ground segmentation for reliable loop closure in degenerate urban environments (ICRA 2022 / IJRR 2023).

SELECTED PROJECTS

- **[e.g., Real-Time LiDAR SLAM System]** – Reduced end-to-end localization drift by X% in city-scale environments by implementing degeneracy-aware factor graph optimization, running at 10 Hz on embedded hardware ([Team/Company, Year])
- **[e.g., 3D Object Detection Pipeline]** – Improved long-range detection recall by X% by fusing LiDAR and camera with a late-fusion Transformer, processing at 20 Hz on NVIDIA Orin ([Team/Company, Year])
- **[e.g., Incremental HD Map Builder]** – Scaled occupancy mapping to campus-scale (>1 km²) by designing an incremental voxel compression pipeline, cutting storage by 60% vs. naive accumulation ([Team/Company, Year])

PUBLICATIONS

10+ first-author papers at top-tier robotics and AI venues: IEEE ICRA, RSS, IEEE RA-L, NeurIPS, IEEE/CVF ICCV, IJRR

AWARDS

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| 2025 | Outstanding Reviewer Award, IEEE ICRA'25 (5 awardees out of 7,641 reviewers) |
| 2024 | RSS Pioneers 2024 (30 selected out of 202 candidates) |
| 2024 | 1st Prize, HILTI SLAM Challenge, IEEE ICRA'24 |
| 2023 | Best Paper Award, IEEE RA-L (5 of 1,100 papers) |
| 2023 | 1st Prize, HILTI SLAM Challenge, IEEE ICRA'23 (63 international teams) |
| 2023 | CES'23 Innovation Award (via technology transfer with HILLS Robotics) |
| 2022 | 2nd Prize, HILTI SLAM Challenge, IEEE ICRA'22 |
| 2020 | Student Best Paper Award, ICCAS |